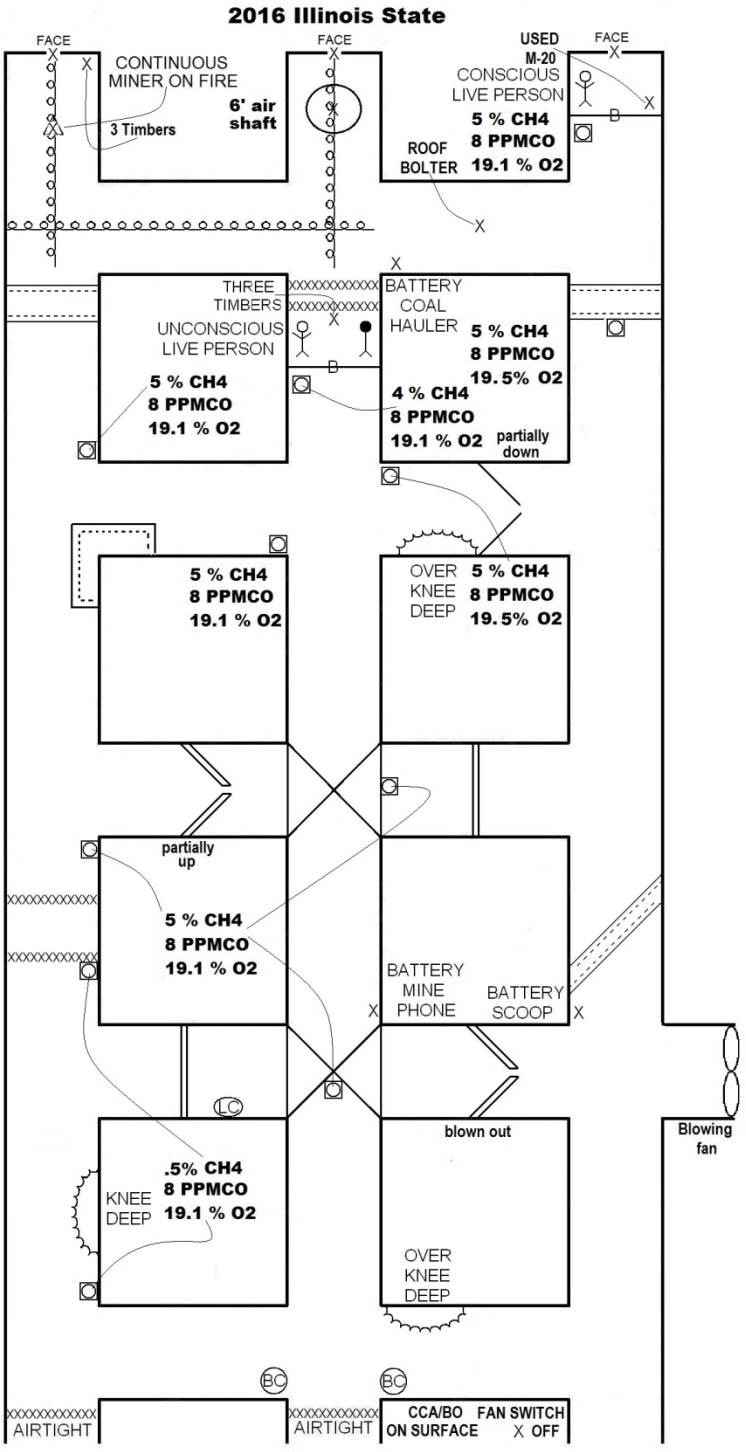
2016 Illinois State Mine Rescue Contest



Problem Briefing

August 24th 2016

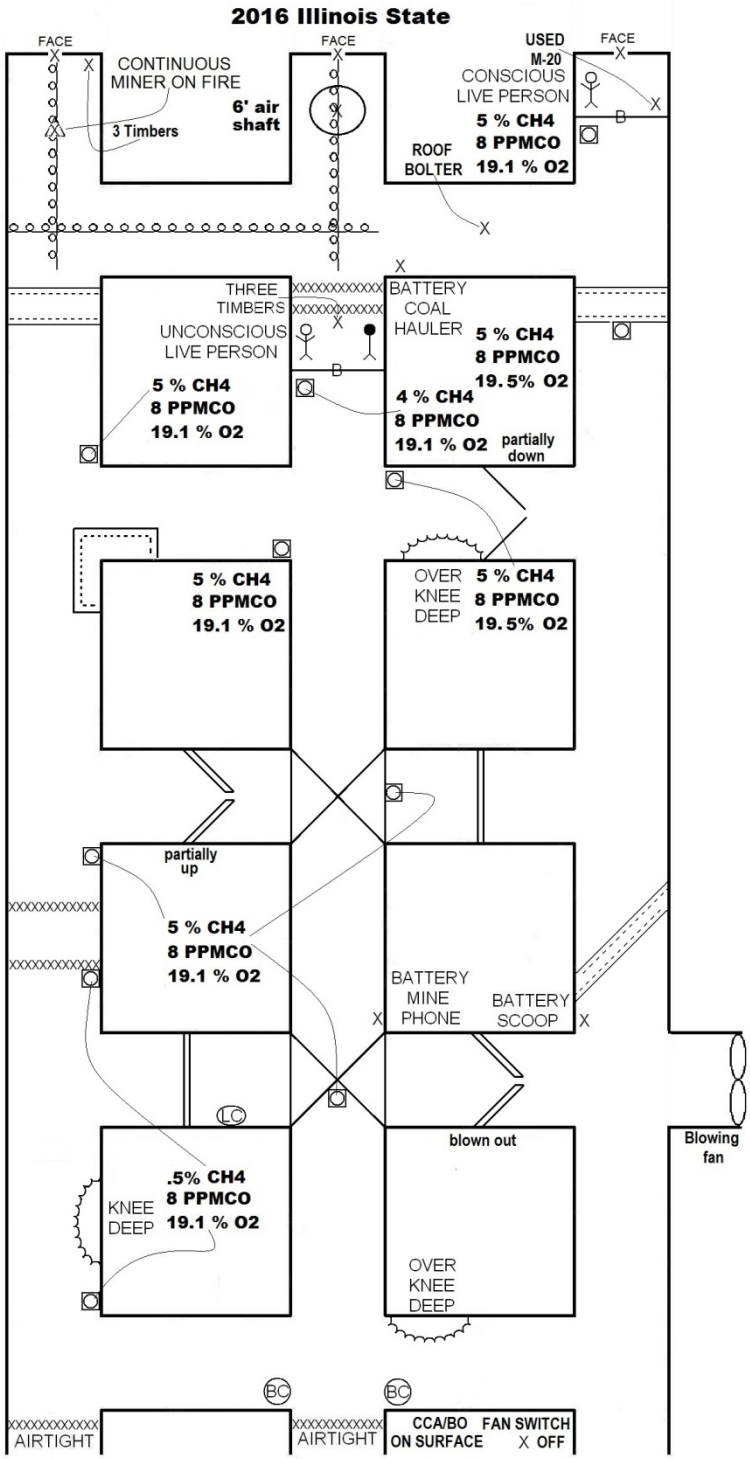

2016 Illinois State
FACE
FACE
FACE
CONTINUOUS
MINER ON FIRE
6' air
shaft
3 Timbers
USED
M-20
CONSCIOUS
LIVE PERSON
5 % CH4
8 PPMCO
19.1 % O2
ROOF
BOLTER
THREE
TIMBERS
UNCONSCIOUS
LIVE PERSON
5 % CH4
8 PPMCO
19.1 % O2
BATTERY
COAL
HAULER
5 % CH4
8 PPMCO
19.5% O2
4 % CH4
8 PPMCO
19.1 % O2
partially
down
5 % CH4
8 PPMCO
19.1 % O2
OVER
KNEE
DEEP
5 % CH4
8 PPMCO
19.5% O2
partially
up
5 % CH4
8 PPMCO
19.1 % O2
BATTERY
MINE
PHONE
BATTERY
SCOOP
LC
.5% CH4
8 PPMCO
19.1 % O2
KNEE
DEEP
blown out
OVER
KNEE
DEEP
Blowing
fan
BC
BC
AIRTIGHT
AIRTIGHT
CCA/BO
ON SURFACE
FAN SWITCH
X OFF

Problem Statement

A fresh air base has been established in the first crosscut of the Blue Blaze no. 2 mine. The mine is ventilated by a blowing fan, which is off at the present time, but can be started by a fan switch in the command center. An airshaft is located in the no. 2 entry and can be used for ventilation if needed by your rescue team during exploration.

Early this morning, a 2 man maintenance crew along with their foreman entered the mine to repair a continuous miner located in the # 1 entry. A call from the section stated that there was a large inundation of gas, that the crew was feeling dis-oriented, and they needed immediate assistance. Shortly after the call, an explosion was heard and the fan stopped. This was the last communication we had with the crew.

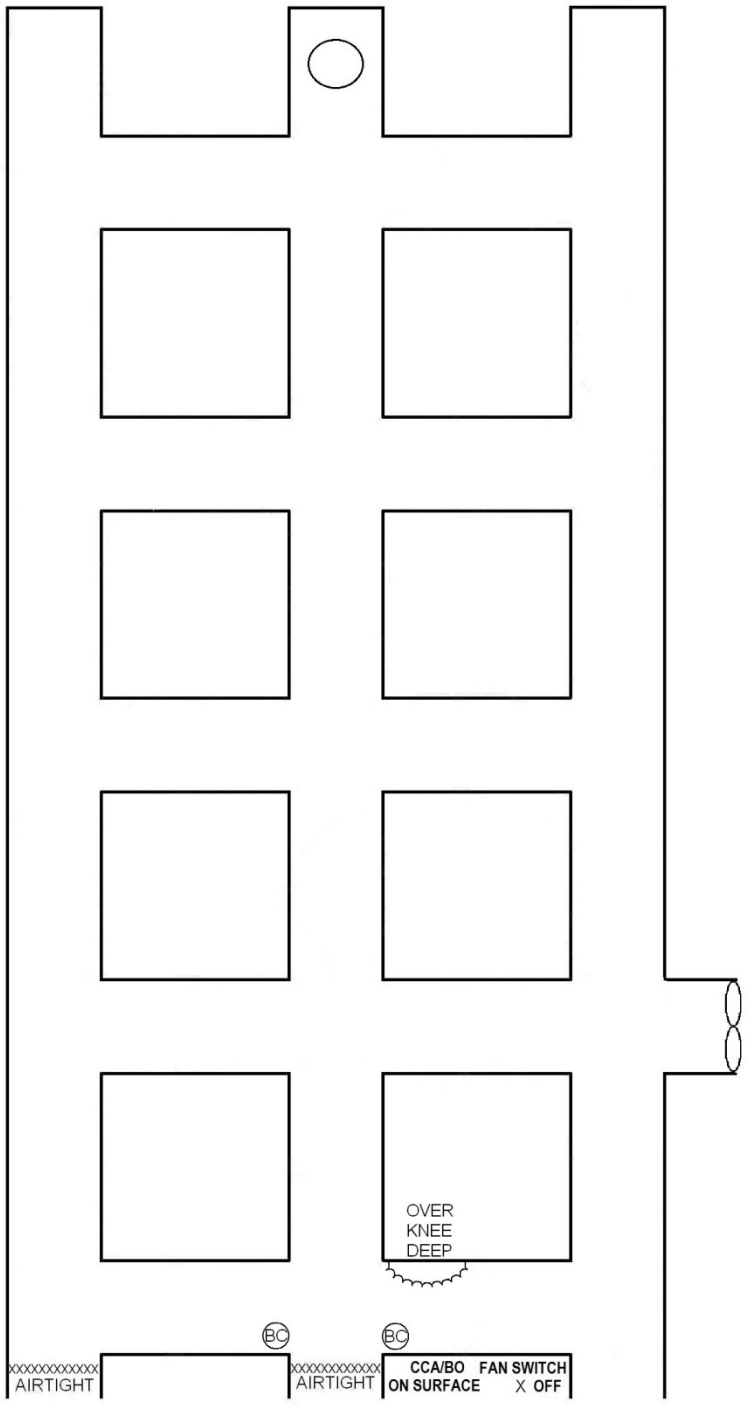
We have de-energized all electrical power inby the fresh air base. All the federal and state agencies have been notified and are on site to offer assistance. The maps are up to date. This mine is noted for poor roof, high liberations of methane, and low oxygen levels. Water accumulations are also frequently found. A fully equipped back up mine rescue team is stationed here while you work inby the fresh air base. You may start work when your team is ready. Good luck, and thank you for responding to our emergency.

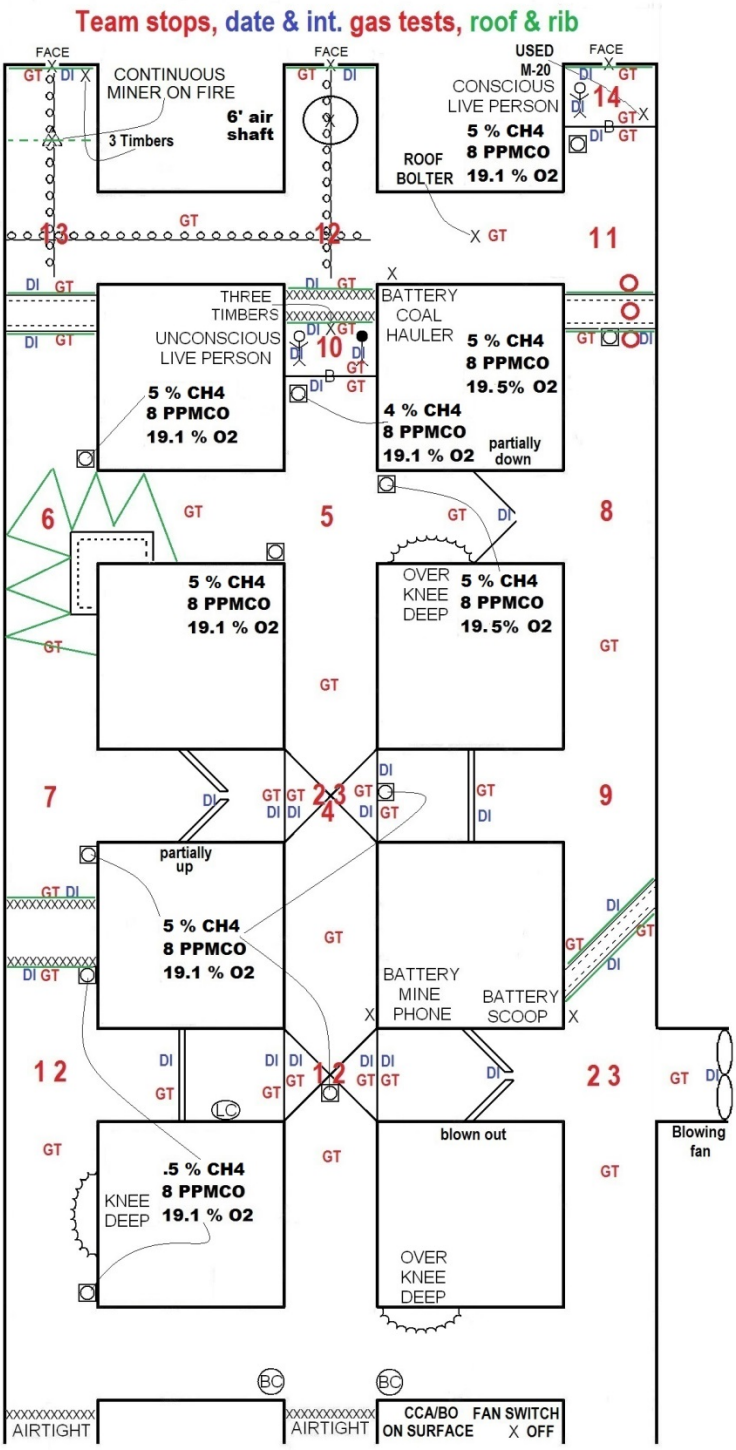

2016 Illinois State
FACE
FACE
FACE
CONTINUOUS MINER ON FIRE
6' air shaft
3 Timbers
USED M-20
CONSCIOUS LIVE PERSON
5 % CH4
8 PPMCO
19.1 % O2
ROOF BOLTER
THREE TIMBERS
UNCONSCIOUS LIVE PERSON
5 % CH4
8 PPMCO
19.1 % O2
BATTERY COAL HAULER
5 % CH4
8 PPMCO
19.5% O2
4 % CH4
8 PPMCO
19.1 % O2
partially down
5 % CH4
8 PPMCO
19.1 % O2
OVER KNEE DEEP
5 % CH4
8 PPMCO
19.5% O2
partially up
5 % CH4
8 PPMCO
19.1 % O2
BATTERY MINE PHONE
BATTERY SCOOP
LC
.5% CH4
8 PPMCO
19.1 % O2
KNEE DEEP
blown out
OVER KNEE DEEP
Blowing fan
BC
BC
AIRTIGHT
AIRTIGHT
CCA/BO ON SURFACE
FAN SWITCH X OFF

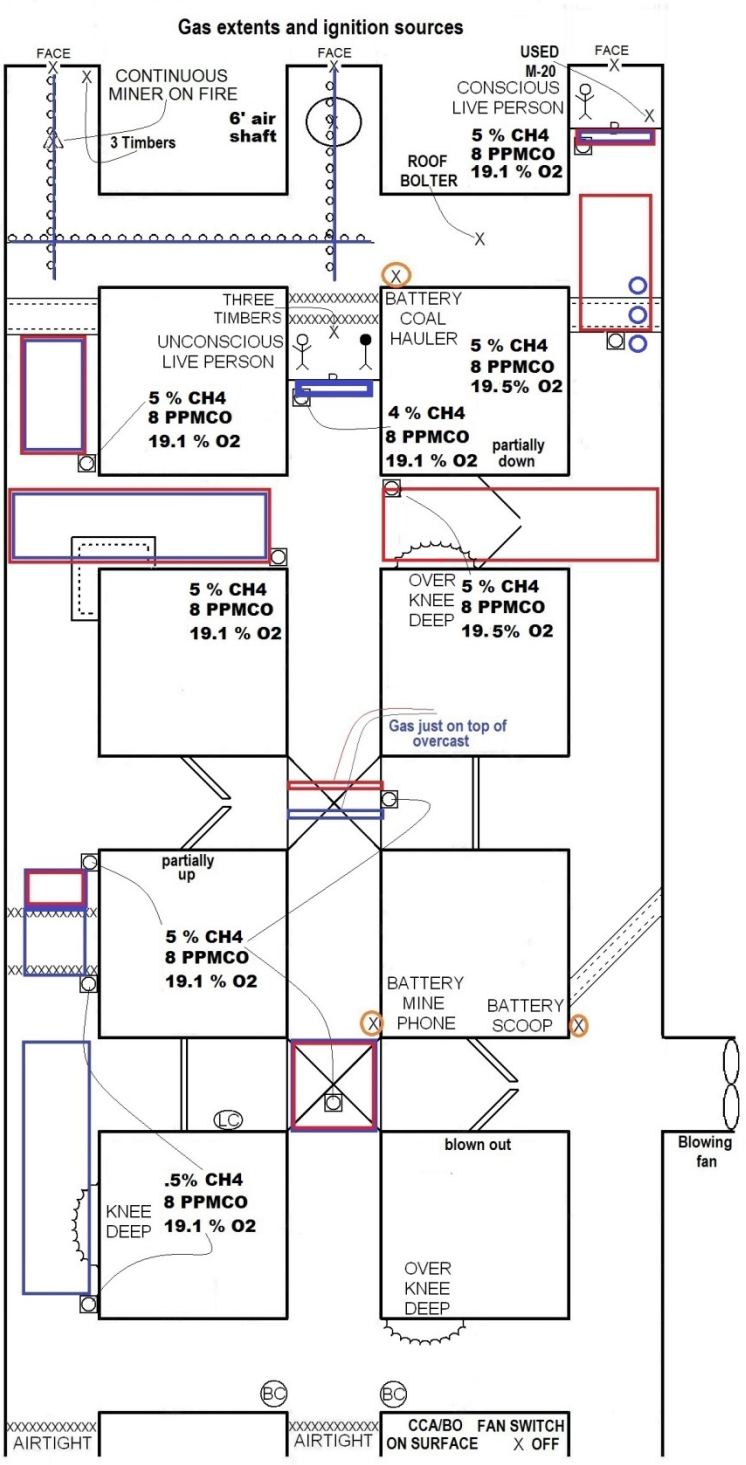
2016 Illinois State Mine Rescue Contest

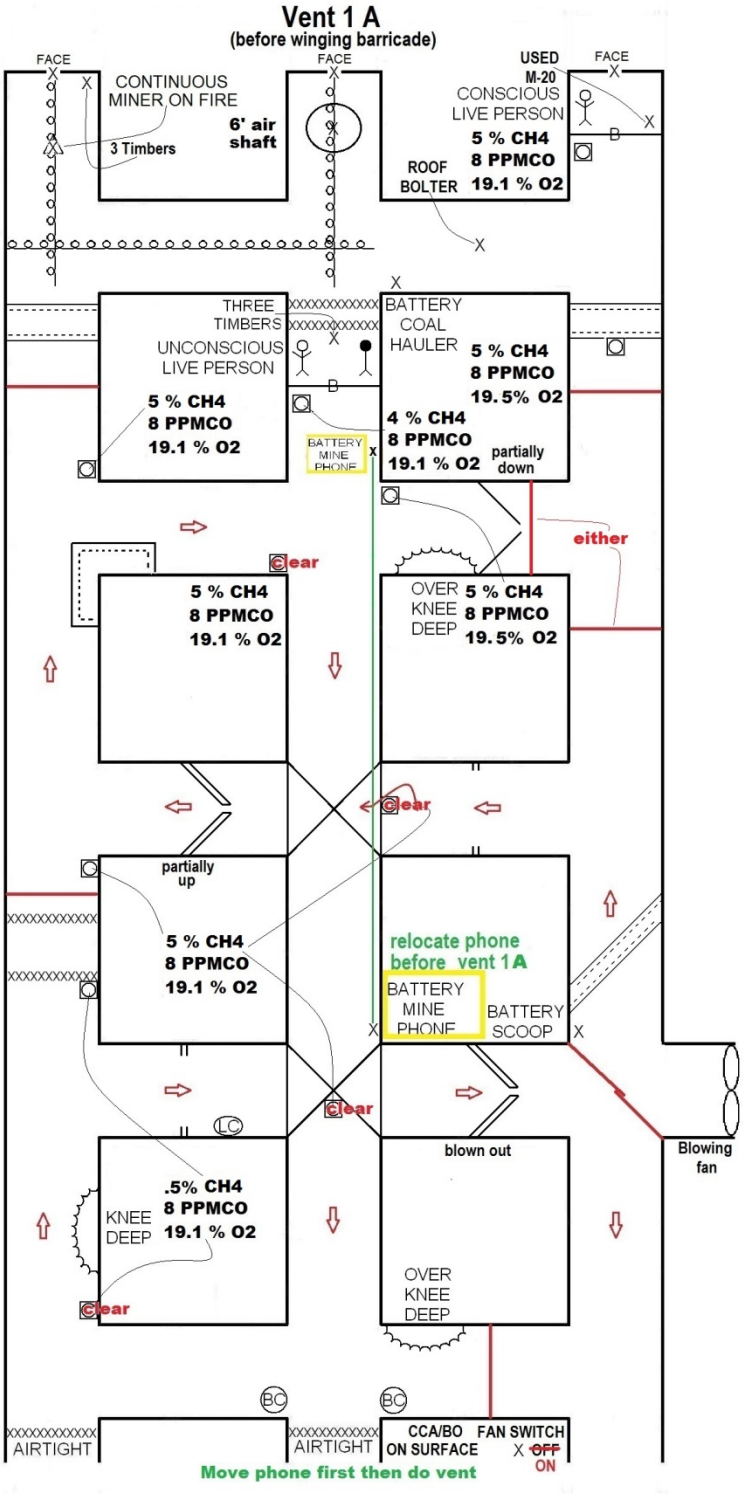
Written Instructions

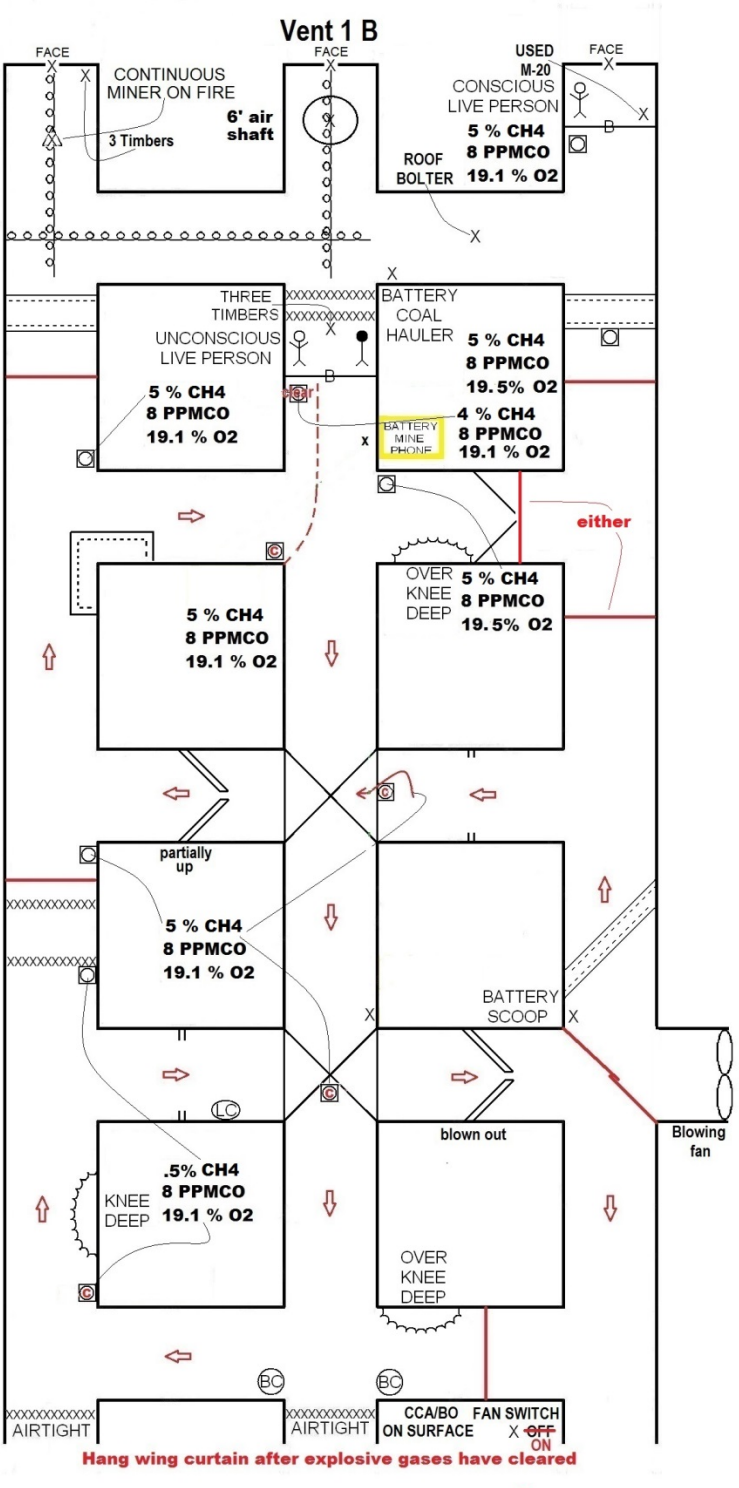
- **ACCOUNT FOR ALL MISSING MINERS**
- **THE BLOWING FAN CAN BE TURNED ON AND OFF BY A SWITCH IN THE COMMAND CENTER**
- **THE BLOWING FAN CAN NEVER BE REVERSED AND CANNOT BE STALLED WHILE IN OPERATION**
- **NO WORK CAN BE PREFORMED BY THE BACK UP TEAM**


OVER
KNEE
DEEP
BC
BC
AIRTIGHT
AIRTIGHT
CCA/BO FAN SWITCH
ON SURFACE X OFF


Team stops, date & int. gas tests, roof & rib
FACE
CONTINUOUS MINER ON FIRE
6' air shaft
3 Timbers
USED M-20
CONSCIOUS LIVE PERSON
5 % CH4
8 PPMCO
19.1 % O2
ROOF BOLTER
THREE TIMBERS
UNCONSCIOUS LIVE PERSON
BATTERY COAL HAULER
5 % CH4
8 PPMCO
19.5% O2
4 % CH4
8 PPMCO
19.1 % O2
partially down
OVER KNEE DEEP
partially up
BATTERY MINE PHONE
BATTERY SCOOP
blown out
.5 % CH4
8 PPMCO
19.1 % O2
KNEE DEEP
Blowing fan
AIRTIGHT
AIRTIGHT
CCA/BO ON SURFACE
FAN SWITCH X OFF

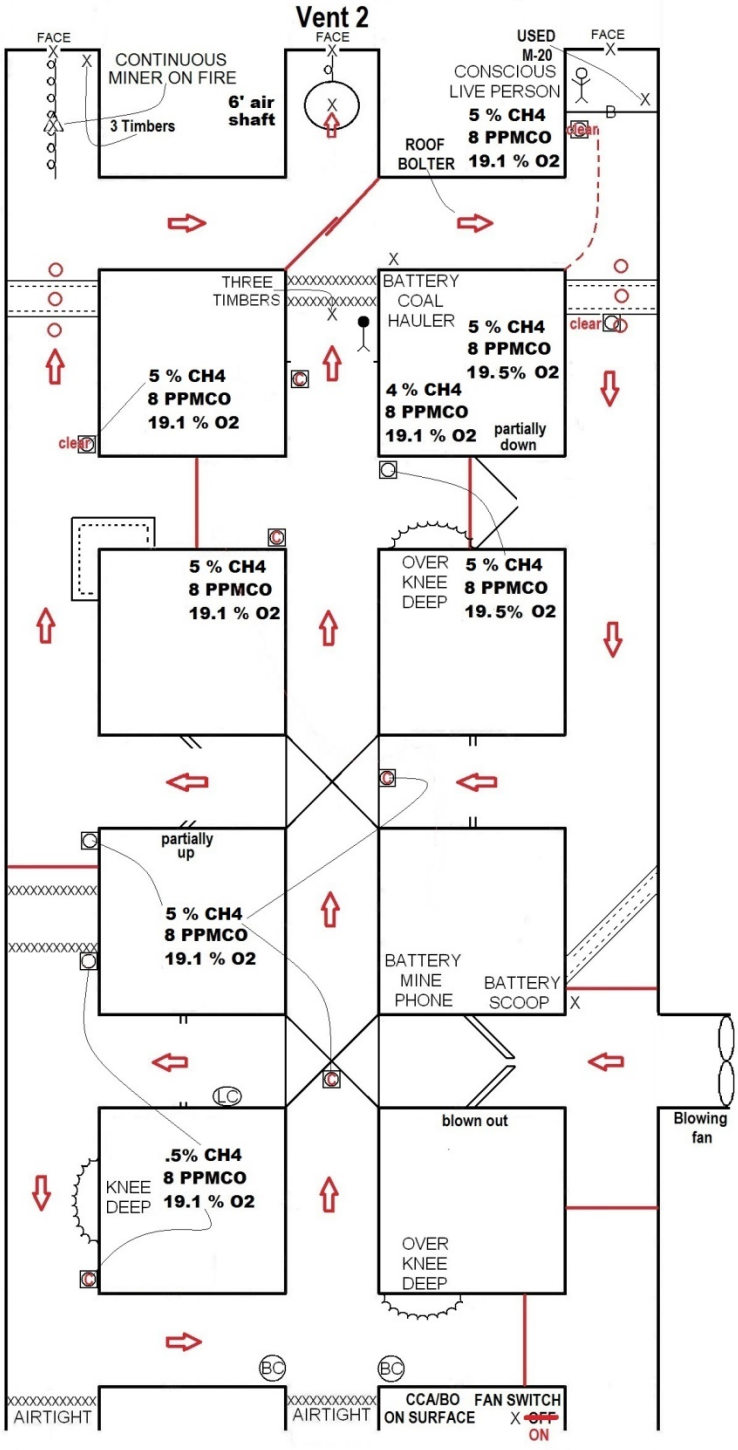

Gas extents and ignition sources
FACE
FACE
FACE
CONTINUOUS MINER ON FIRE
6' air shaft
3 Timbers
USED M-20
CONSCIOUS LIVE PERSON
5 % CH4
8 PPMCO
19.1 % O2
ROOF BOLTER
THREE TIMBERS
UNCONSCIOUS LIVE PERSON
5 % CH4
8 PPMCO
19.1 % O2
BATTERY COAL HAULER
5 % CH4
8 PPMCO
19.5% O2
4 % CH4
8 PPMCO
19.1 % O2
partially down
5 % CH4
8 PPMCO
19.1 % O2
OVER KNEE DEEP
5 % CH4
8 PPMCO
19.5% O2
Gas just on top of overcast
partially up
5 % CH4
8 PPMCO
19.1 % O2
BATTERY MINE PHONE
BATTERY SCOOP
LC
.5% CH4
8 PPMCO
19.1 % O2
KNEE DEEP
blown out
OVER KNEE DEEP
Blowing fan
BC
BC
AIRTIGHT
AIRTIGHT
CCA/BO ON SURFACE
FAN SWITCH X OFF

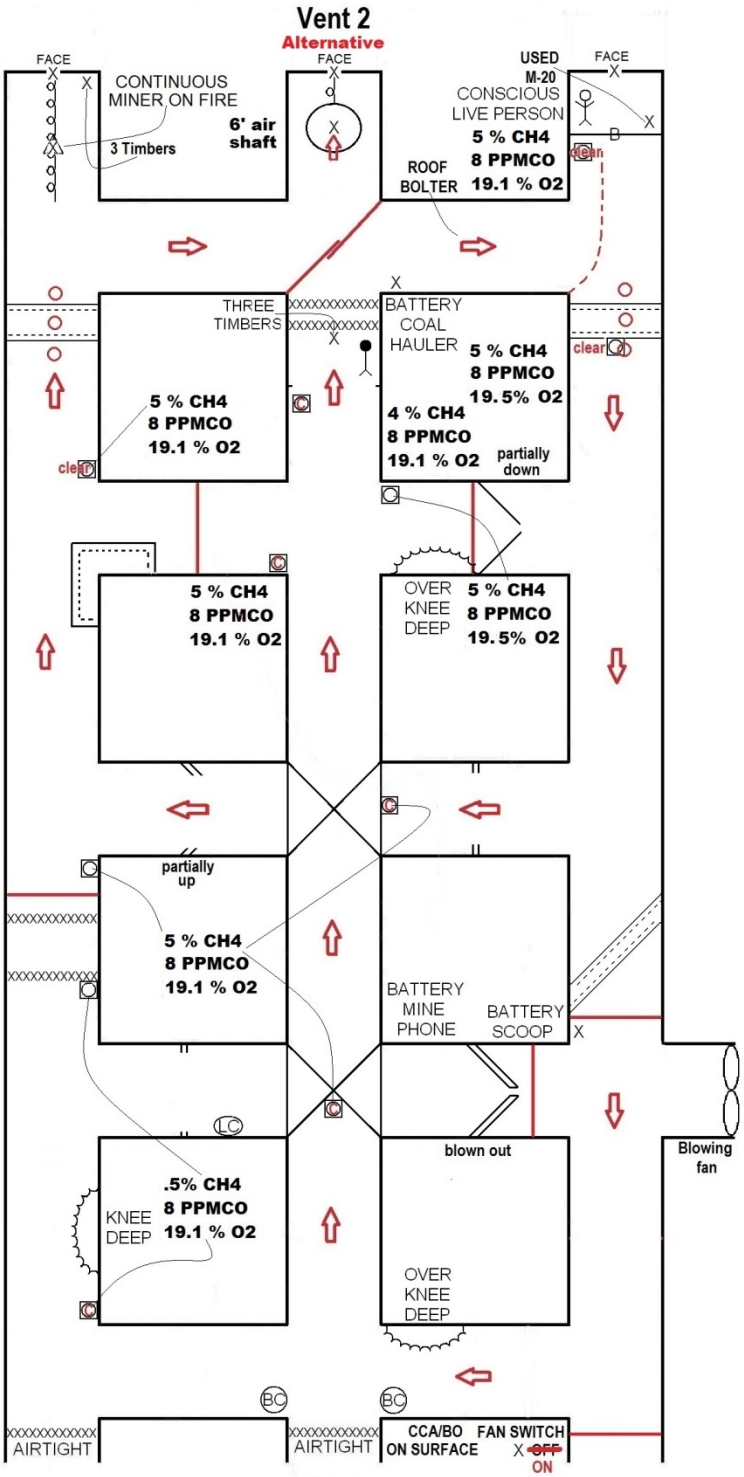

Vent 1 A
(before winging barricade)
FACE
FACE
FACE
CONTINUOUS
MINER ON FIRE
6' air
shaft
3 Timbers
USED
M-20
CONSCIOUS
LIVE PERSON
5 % CH4
8 PPMCO
19.1 % O2
ROOF
BOLTER
THREE
TIMBERS
UNCONSCIOUS
LIVE PERSON
5 % CH4
8 PPMCO
19.1 % O2
BATTERY
COAL
HAULER
5 % CH4
8 PPMCO
19.5% O2
4 % CH4
8 PPMCO
19.1 % O2
partially
down
BATTERY
MINE
PHONE
either
Clear
5 % CH4
8 PPMCO
19.1 % O2
OVER
KNEE
DEEP
5 % CH4
8 PPMCO
19.5% O2
Clear
partially
up
5 % CH4
8 PPMCO
19.1 % O2
relocate phone
before vent 1A
BATTERY
MINE
PHONE
BATTERY
SCOOP
Clear
LC
blown out
Blowing
fan
.5% CH4
8 PPMCO
19.1 % O2
KNEE
DEEP
Clear
OVER
KNEE
DEEP
BC
BC
AIRTIGHT
AIRTIGHT
CCA/BO
ON SURFACE
FAN SWITCH
OFF
ON
Move phone first then do vent


Vent 1 B
FACE
FACE
FACE
CONTINUOUS MINER ON FIRE
6' air shaft
3 Timbers
USED M-20
CONSCIOUS LIVE PERSON
5 % CH4
8 PPMCO
19.1 % O2
ROOF BOLTER
THREE TIMBERS
UNCONSCIOUS LIVE PERSON
5 % CH4
8 PPMCO
19.1 % O2
BATTERY COAL HAULER
5 % CH4
8 PPMCO
19.5% O2
BATTERY MINE PHONE
4 % CH4
8 PPMCO
19.1 % O2
either
5 % CH4
8 PPMCO
19.1 % O2
OVER KNEE DEEP
5 % CH4
8 PPMCO
19.5% O2
partially up
5 % CH4
8 PPMCO
19.1 % O2
BATTERY SCOOP
LC
blown out
Blowing fan
.5% CH4
8 PPMCO
19.1 % O2
KNEE DEEP
OVER KNEE DEEP
BC
BC
AIRTIGHT
AIRTIGHT
CCA/BO ON SURFACE
FAN SWITCH
OFF
ON
Hang wing curtain after explosive gases have cleared

PATIENT STATEMENT IN BARRICADE IN NO. 3 ENTRY INBY LAST OPEN

HELP! IT'S AIRTIGHT BEHIND ME……


Vent 2
FACE
CONTINUOUS MINER ON FIRE
3 Timbers
6' air shaft
FACE
USED M-20
CONSCIOUS LIVE PERSON
5 % CH4
8 PPMCO
19.1 % O2
FACE
ROOF BOLTER
THREE TIMBERS
BATTERY COAL HAULER
5 % CH4
8 PPMCO
19.5% O2
4 % CH4
8 PPMCO
19.1 % O2
partially down
clear
5 % CH4
8 PPMCO
19.1 % O2
OVER KNEE DEEP
5 % CH4
8 PPMCO
19.5% O2
partially up
5 % CH4
8 PPMCO
19.1 % O2
BATTERY MINE PHONE
BATTERY SCOOP
LC
blown out
Blowing fan
.5% CH4
8 PPMCO
19.1 % O2
KNEE DEEP
OVER KNEE DEEP
BC
BC
AIRTIGHT
AIRTIGHT
CCA/BO ON SURFACE
FAN SWITCH
OFF
ON


Vent 2
Alternative
FACE
CONTINUOUS MINER ON FIRE
6' air shaft
3 Timbers
FACE
USED M-20
CONSCIOUS LIVE PERSON
5 % CH4
8 PPMCO
19.1 % O2
FACE
ROOF BOLTER
THREE TIMBERS
BATTERY COAL HAULER
5 % CH4
8 PPMCO
19. 5% O2
4 % CH4
8 PPMCO
19.1 % O2
partially down
clear
5 % CH4
8 PPMCO
19.1 % O2
OVER KNEE DEEP
5 % CH4
8 PPMCO
19. 5% O2
partially up
5 % CH4
8 PPMCO
19.1 % O2
BATTERY MINE PHONE
BATTERY SCOOP
LC
.5% CH4
8 PPMCO
19.1 % O2
KNEE DEEP
blown out
OVER KNEE DEEP
Blowing fan
BC
BC
AIRTIGHT
AIRTIGHT
CCA/BO ON SURFACE
FAN SWITCH
OFF
ON

Keys to Problem

- **Blowing fan is inby fresh air base and will create air migration after turning on**
- **Team must follow contaminated entry while exploring (See team stop map)**
- **Team must explore behind intact permanent stoppings before advancing the no.5 man inby the corner of the second crosscut**
- **Team must move battery mine phone inby before first ventilation (See vent 1A map)**
- **Team must not send explosive mixture over battery mine phone or other battery equipment**
- **Team must not send explosive mixtures over unexplored areas and not send low O2 over unexplored area until all persons are accounted for**
- **Team must timber unsafe roof in no.3 entry to follow contaminate into last open crosscut**
- **Team must timber unsafe roof in no.1 entry to complete 2nd vent (See vent 2 map)**